BATCH 11

Smart Hostel Energy Saver System

Automatic room power cutoff using PIR (Passive Infrared) heat detection when no occupant is present — saving energy, reducing hostel bills.

ESP32 · HC-SR501 PIR · Relay Module · Arduino IDE

THE PROBLEM

28-30%

of hostel electricity
wasted in empty rooms

₹800+

extra per student per year
due to forgotten lights/fans

~0

existing hostels with
auto power-off systems

Students routinely leave lights and fans running when they step out — lectures, canteen, sports. In a 200-room hostel, this adds up to significant electricity waste every single day.

OUR SOLUTION

If nobody is in the room, cut the power. Automatically.

PIR Heat Detection

Senses infrared radiation from human body heat — no contact needed

Relay Auto Switch

Physically cuts power to lights & fan via relay when room is empty

5-Min Timeout

Waits 5 minutes after last motion before switching off — no false cuts

Manual Override

Students can override with a button for uninterrupted study time

COMPONENTS & CIRCUIT

ESP32 DevKit C V4

Main microcontroller — reads PIR, controls relay, runs logic

HC-SR501 PIR Sensor

Detects body heat (infrared) — GPIO 13, range 7m, 120°

Relay Module

Physically switches AC power to room appliances — GPIO 12

LDR Sensor

Ambient light check — only activates at night via GPIO 34 (ADC)

Push Button

Manual 30-min override for continuous use — GPIO 35

LED Indicator

Visual status — ON when relay is active, 220Ω resistor

🔗 Simulated on Wokwi — wokwi.com/projects/456377440609481729

HOW IT WORKS

1

PIR Scans Room

Constantly detects infrared heat signatures from human bodies

2

Motion Detected

ESP32 reads HIGH signal → timer resets → relay stays ON

3

Room Empty?

No heat detected for 5 continuous minutes

4

Power Cut

Relay switches OFF → lights and fan turn off automatically

```
if (motion == HIGH && isDark) { lastMotionTime = millis(); setRelay(true); }  //
5-min timeout auto-OFF
```

FUTURE IMPROVEMENTS

Phase 1

- PIR heat detection
- Relay auto switch
- 5-min timeout
- LED indicator

Phase 2

- LDR ambient light check
- Manual override button
- 2-channel relay (fan + light)
- OLED status display

Phase 3

- WiFi dashboard (ESP32)
- Energy usage analytics
- Warden mobile alerts
- Central hostel control

COST ESTIMATE

Component	Qty	Unit Price	Total
ESP32 DevKit C V4	1	₹350	₹350
HC-SR501 PIR Sensor	1	₹65	₹65
5V Relay Module	1	₹55	₹55
LDR + 10kΩ Resistor	1	₹15	₹15
Push Button	1	₹5	₹5
220Ω Resistors	2	₹2	₹4
LEDs	2	₹5	₹10
Breadboard (830pt)	1	₹70	₹70
Jumper Wires	1	₹50	₹50
USB Cable	1	₹80	₹80
PROTOTYPE TOTAL			**₹704**

Production Unit (Per Room)

PCB Fabrication **₹250**

Project Enclosure **₹80**

5V Wall Adapter **₹120**

Total per Room: ₹1,150

💰 ROI for 100-room hostel

Deploy cost: ~₹1.15 lakh

Est. monthly savings: ₹8,000+

Break-even in < 15 months

CONCLUSION

A Practical, Low-Cost Solution for Hostel Energy Management

🌿 Scientifically grounded — PIR genuinely detects human body heat

💰 Cost-effective — ₹1,150/room with fast ROI at hostel scale

🚀 Expandable — WiFi, analytics, and multi-room control ready

Simulated on Wokwi · Ready for Prototype Build